

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

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LAB REPORT

on

OPERATING SYSTEMS

Submitted by

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in partial fulfillment for the award of the degree of
BACHELOR OF ENGINEERING
in
COMPUTER SCIENCE AND ENGINEERING



B.M.S. COLLEGE OF ENGINEERING

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CERTIFICATE

This is to certify that the Lab work entitled “OPERATING SYSTEMS – 23CS4PCOPS” carried out by A Praful Srinivasan (1BM24CS001), who is Bonafide student of B. M. S. College of Engineering. It is in partial fulfilment for the award of Bachelor of Engineering in Computer Science and Engineering of the Visvesvaraya Technological University, Belgaum during the year Feb-2026 to June-2026.
The Lab report has been approved as it satisfies the academic requirements in respect of a OPERATING SYSTEMS - (23CS4PCOPS) work prescribed for the said degree.

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2.	Write a C program to simulate multi-level queue scheduling algorithm considering the following scenario. All the processes in the system are divided into two categories – system processes and user processes. System processes are to be given higher priority than user processes. Use FCFS scheduling for the processes in each queue.	
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Course Outcomes

C01	Apply the different concepts and functionalities of Operating System
C02	Analyse various Operating system strategies and techniques
C03	Demonstrate the different functionalities of Operating System.
C04	Conduct practical experiments to implement the functionalities of Operating system.

Program1a

Question: Write a C program to simulate the following non-pre-emptive CPU scheduling algorithm to find turnaround time and waiting time.

- FCFS
- SJF (pre-emptive & Non-preemptive)

Soln: =>FCFS

```
#include <stdio.h>
#include <stdlib.h>

struct process {
    int id, at, bt, ct, tat, wt, rt;
};

int main() {
    int n;
    printf("Enter the number of processes: ");
    scanf("%d", &n);

    struct process p[n];
    for (int i = 0; i < n; i++) {
        p[i].id = i + 1;
        printf("Enter Arrival Time for P%d: ", p[i].id);
        scanf("%d", &p[i].at);
        printf("Enter Burst Time for P%d: ", p[i].id);
        scanf("%d", &p[i].bt);
    }
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n - i - 1; j++) {
            if (p[j].at > p[j + 1].at) {
                struct process temp = p[j];
                p[j] = p[j + 1];
                p[j + 1] = temp;
            }
        }
    }
    int timepassed = 0, sumtat = 0, sumwt = 0, sumrt = 0;
    for (int i = 0; i < n; i++) {
        if (timepassed < p[i].at) {
            timepassed = p[i].at;
        }
        p[i].ct = timepassed + p[i].bt;
        p[i].tat = p[i].ct - p[i].at;
        p[i].wt = p[i].tat - p[i].bt;

        timepassed += p[i].bt;

        sumtat += p[i].tat;
        sumwt += p[i].wt;
    }
    printf("\nID\tAT\tBT\tCT\tTAT\tWT\tRT\n");
```

```

for (int i = 0; i < n; i++) {
    printf("%d\t%d\t%d\t%d\t%d\t%d\t%d\n",
        p[i].id, p[i].at, p[i].bt, p[i].ct,
        p[i].tat, p[i].wt);
}
printf("\nAverage TAT: %.2f\n", (float)sumtat / n);
printf("Average WT: %.2f\n", (float)sumwt / n);

return 0;
}

```

```

C:\Users\BMSCECSE-L4\Desktop >
Enter the number of processes
4
Arrival Time for Process P10
Burst Time for process P12
Arrival Time for Process P21
Burst Time for process P22
Arrival Time for Process P35
Burst Time for process P33
Arrival Time for Process P46
Burst Time for process P44
ProcessID      Arrival Time      Burst Time      Completion Time  Turnaround Time  Waiting Time
1              0                2              2                2                0
2              1                2              4                3                1
3              5                3              8                3                0
4              6                4              12               6                2
Average TAT:3.50
Average WT:0.75
DONE BY A PRAFUL SRINIVASAN

Process returned 0 (0x0)   execution time : 25.796 s
Press any key to continue.
|

```

SOLN: =>SJF(Preemptive)

```
#include<stdio.h>
```

```
struct process {  
    int id, at, bt, ct, tat, wt, rt, rem;  
};
```

```
int main() {  
    int n;  
    printf("Enter the number of processes: ");  
    scanf("%d", &n);
```

```
    struct process p[n];  
    for (int i = 0; i < n; i++) {  
        p[i].id = i + 1;  
        printf("Enter Arrival Time for P%d: ", p[i].id);  
        scanf("%d", &p[i].at);  
        printf("Enter Burst Time for P%d: ", p[i].id);  
        scanf("%d", &p[i].bt);  
        p[i].rem = p[i].bt;  
        p[i].rt = -1;  
    }
```

```
    int completed = 0, time = 0, sumtat = 0, sumwt = 0, sumrt = 0;  
    while (completed != n) {  
        int idx = -1, minrem = 1e9;  
        for (int i = 0; i < n; i++) {  
            if (p[i].at <= time && p[i].rem > 0) {  
                if (p[i].rem < minrem || (p[i].rem == minrem && p[i].at < p[idx].at)) {  
                    minrem = p[i].rem;  
                    idx = i;  
                }  
            }  
        }  
        if (idx == -1) {  
            time++;  
            continue;  
        }  
        if (p[idx].rt == -1) {  
            p[idx].rt = time - p[idx].at;  
            sumrt += p[idx].rt;  
        }  
        p[idx].rem--;  
        time++;  
        if (p[idx].rem == 0) {  
            completed++;  
            p[idx].ct = time;  
            p[idx].tat = p[idx].ct - p[idx].at;  
            p[idx].wt = p[idx].tat - p[idx].bt;  
            sumtat += p[idx].tat;  
            sumwt += p[idx].wt;  
        }  
    }
```

```

printf("\nID\tAT\tBT\tCT\tTAT\tWT\n");
for (int i = 0; i < n; i++) {
    printf("%d\t%d\t%d\t%d\t%d\t%d\n",
        p[i].id, p[i].at, p[i].bt, p[i].ct,
        p[i].tat, p[i].wt);
}

printf("\nAverage TAT: %.2f\n", (float)sumtat / n);
printf("Average WT: %.2f\n", (float)sumwt / n);
return 0;
}

```

```

C:\Users\BMSCECSE-L4\Desktop >
Enter the number of processes5
Arrival Time of P12
Burst time of P11
Arrival Time of P21
Burst time of P25
Arrival Time of P34
Burst time of P31
Arrival Time of P40
Burst time of P46
Arrival Time of P52
Burst time of P53
ProcessID      Arrival Time      Burst Time      Completion Time  Turnaround Time  Waiting Time
1              2                1               3                1                0
3              4                1               5                1                0
5              2                3               7                5                2
2              1                5               11               10               5
4              0                6               16               16               10
Average TAT: 6.60
Average WT:3.40
DONE BY A PRAFUL SRINIVASAN

Process returned 0 (0x0)   execution time : 27.752 s
Press any key to continue.
|

```

SOLN: => SJF Non preemptive

```
#include<stdio.h>
struct process{
    int id,at,bt,ct,tat,wt,rt;
};

int main() {
    int n;
    printf("Enter the number of processes: ");
    scanf("%d", &n);

    struct process p[n];
    for (int i = 0; i < n; i++){
        p[i].id = i + 1;
        printf("Enter Arrival Time for P%d: ", p[i].id);
        scanf("%d", &p[i].at);
        printf("Enter Burst Time for P%d: ", p[i].id);
        scanf("%d", &p[i].bt);
    }

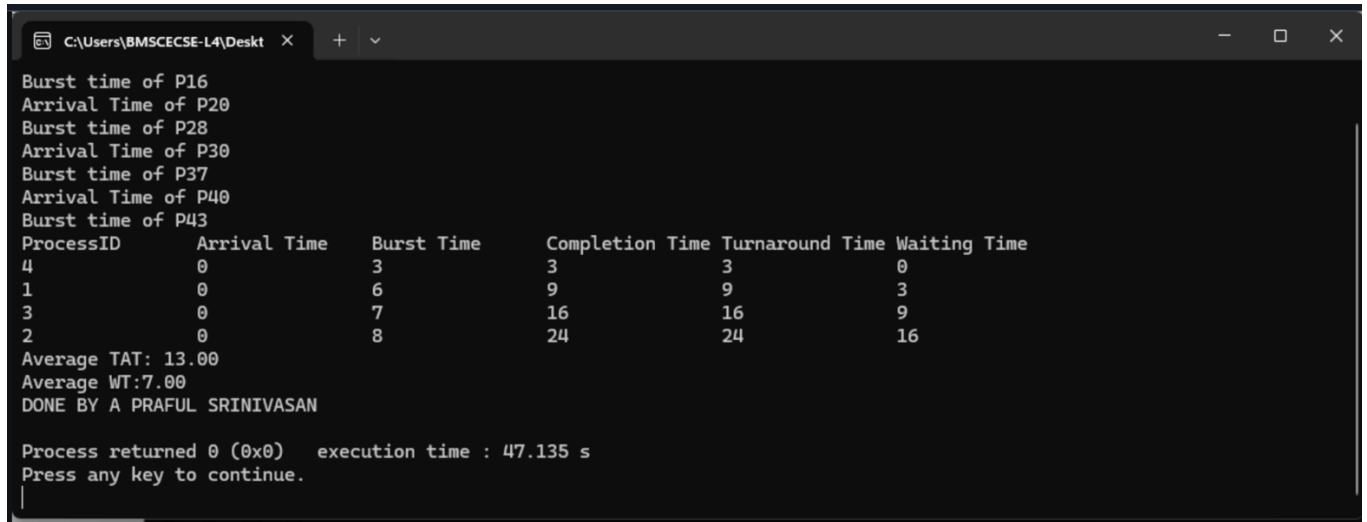
    int completed=0,time=0,sumtat=0,sumwt=0;
    int iscompleted[n];
    for(int i=0;i<n;i++) iscompleted[i]=0;

    while(completed!=n){
        int index=-1;
        int minbt=1e9;
        for(int i=0;i<n;i++){
            if(p[i].at<=time && !iscompleted[i]){
                if(p[i].bt<minbt){
                    minbt=p[i].bt;
                    index=i;
                }
            }
        }
        if(index!=-1){
            p[index].ct=time+p[index].bt;
            p[index].tat=p[index].ct-p[index].at;
            p[index].wt=p[index].tat-p[index].bt;
            p[index].rt=p[index].wt;
            sumtat+=p[index].tat;
            sumwt+=p[index].wt;
            time=p[index].ct;
            iscompleted[index]=1;
            completed++;
        } else {
            time++;
        }
    }

    printf("\nID\tAT\tBT\tCT\tTAT\tWT\n");
    for(int i=0;i<n;i++){
        printf("%d\t%d\t%d\t%d\t%d\t%d\n",p[i].id,p[i].at,p[i].bt,p[i].ct,p[i].tat,p[i].wt);
    }
}
```



```
printf("\nAverage TAT: %.2f", (float)sumtat/n);  
printf("\nAverage WT: %.2f", (float)sumwt/n);  
}
```



```
C:\Users\BMSCECSE-L4\Desktop  
Burst time of P16  
Arrival Time of P20  
Burst time of P28  
Arrival Time of P30  
Burst time of P37  
Arrival Time of P40  
Burst time of P43  
ProcessID      Arrival Time    Burst Time      Completion Time Turnaround Time Waiting Time  
4              0              3              3              3              0  
1              0              6              9              9              3  
3              0              7              16             16             9  
2              0              8              24             24             16  
Average TAT: 13.00  
Average WT:7.00  
DONE BY A PRAFUL SRINIVASAN  
  
Process returned 0 (0x0)   execution time : 47.135 s  
Press any key to continue.  
|
```

PROGRAM 1b

Question : Write a C program to simulate the following CPU scheduling algorithm to find turnaround time and waiting time.

→ Priority (pre-emptive & Non-pre-emptive)

→ Round Robin (Experiment with different quantum sizes for RR algorithm)

SOLN: Priority Pre-emptive

```
#include <stdio.h>

struct process {
    int pid, at, bt, p, rt, ct, wt, tat;
};

int main() {
    int n, time = 0, completed = 0;
    printf("Enter the number of processes: ");
    scanf("%d", &n);
    struct process p[n];

    for (int i = 0; i < n; i++) {
        p[i].pid = i + 1;
        printf("Enter the arrival time, burst time, priority: ");
        scanf("%d%d%d", &p[i].at, &p[i].bt, &p[i].p);
        p[i].rt = p[i].bt;
    }

    while (completed < n) {
        int idx = -1, highest = 9999;
        for (int i = 0; i < n; i++) {
            if (p[i].at <= time && p[i].rt > 0) {
                if (p[i].p < highest) {
                    highest = p[i].p;
                    idx = i;
                }
            }
        }
        if (idx != -1) {
            p[idx].rt--;
            time++;
            if (p[idx].rt == 0) {
                completed++;
                p[idx].ct = time;
                p[idx].tat = p[idx].ct - p[idx].at;
                p[idx].wt = p[idx].tat - p[idx].bt;
            }
        } else {
            time++;
        }
    }

    float totalwt = 0, totaltat = 0;
    printf("\nPid\tAt\tBt\tP\tCt\tTat\tWt\n");
```

```

for (int i = 0; i < n; i++) {
    totalwt += p[i].wt;
    totaltat += p[i].tat;
    printf("%d\t%d\t%d\t%d\t%d\t%d\t%d\n",
        p[i].pid, p[i].at, p[i].bt, p[i].p,
        p[i].ct, p[i].tat, p[i].wt);
}

printf("\nAverage waiting time : %.2f", totalwt / n);
printf("\nAverage turnaround time : %.2f\n", totaltat / n);

return 0;
}

```

```

mac@Prafuls-MacBook-Pro BMS % ./"Priority(PE)"
Enter number of processes :3

Enter Burst Time, Arrival Time and Priority for P1:12
1
3

Enter Burst Time, Arrival Time and Priority for P2:11
0
2

Enter Burst Time, Arrival Time and Priority for P3:5
3
4

```

Process	AT	BT	Priority	CT	TAT	WT
P1	1	12	3	23	22	10
P2	0	11	2	11	11	0
P3	3	5	4	28	25	20

```

Average TAT = 19.33
Average WT = 10.00%

```

SOLN : Non premetive

```
#include <stdio.h>

struct process {
    int id, at, bt, pr, ct, tat, wt, d;
};

int main() {
    int n, t = 0, c = 0, h;
    float tw = 0, tt = 0;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    struct process p[n];

    for (int i = 0; i < n; i++) {
        p[i].id = i + 1;
        printf("Process %d\n", p[i].id);
        printf("Arrival time: ");
        scanf("%d", &p[i].at);
        printf("Burst time: ");
        scanf("%d", &p[i].bt);
        printf("Priority: ");
        scanf("%d", &p[i].pr);
        p[i].d = 0;
    }

    while (c != n) {
        h = -1;
        for (int i = 0; i < n; i++) {
            if (p[i].at <= t && p[i].d == 0) {
                if (h == -1 || p[i].pr < p[h].pr) {
                    h = i;
                }
            }
        }

        if (h == -1) {
            t++;
        } else {
            t += p[h].bt;
            p[h].ct = t;
            p[h].tat = p[h].ct - p[h].at;
            p[h].wt = p[h].tat - p[h].bt;
            tw += p[h].wt;
            tt += p[h].tat;
            p[h].d = 1;
            c++;
        }
    }

    printf("\nProcess\tAT\tBT\tPR\tCT\tTAT\tWT\n");
    for (int i = 0; i < n; i++) {
```

```

        printf("%d\t%d\t%d\t%d\t%d\t%d\t%d\t%d\n",
            p[i].id, p[i].at, p[i].bt, p[i].pr,
            p[i].ct, p[i].tat, p[i].wt);
    }

    printf("\nAverage Turnaround Time: %.2f", tt / n);
    printf("\nAverage Waiting Time: %.2f\n", tw / n);

    return 0;
}

```

```

● mac@Prafuls-MacBook-Pro BMS % ./"Priority(NPE)"
Enter number of processes :4

Enter Burst Time, Arrival Time and Priority for P1:12
0
1

Enter Burst Time, Arrival Time and Priority for P2:10
1
3

Enter Burst Time, Arrival Time and Priority for P3:3
3
10

Enter Burst Time, Arrival Time and Priority for P4:4
5
0

Process AT      BT      Priority      CT      TAT      WT
P1      0      12      1      12      12      0
P2      1      10      3      26      25      15
P3      3      3      10      29      26      23
P4      5      4      0      16      11      7

Average TAT = 18.50
Average WT = 11.25%

```

SOLN : Round Robin

```
#include <stdio.h>
```

```
struct Process {  
    int pid, at, bt, ct, tat, wt, rt;  
};
```

```
int main() {  
    int n, tq;  
    printf("Enter number of processes: ");  
    scanf("%d", &n);
```

```
    printf("Enter Time Quantum: ");  
    scanf("%d", &tq);
```

```
    struct Process p[n];  
    int rem_bt[n];  
    int readyQ[100];  
    int front = 0, rear = 0;
```

```
    printf("Enter arrival times:\n");  
    for(int i=0; i<n; i++) {  
        scanf("%d", &p[i].at);  
        p[i].pid = i+1;  
    }
```

```
    printf("Enter burst times:\n");  
    for(int i=0; i<n; i++) {  
        scanf("%d", &p[i].bt);  
        rem_bt[i] = p[i].bt;  
        p[i].rt = p[i].bt;  
        p[i].ct = p[i].tat = p[i].wt = 0;  
    }
```

```
    int time = 0, completed = 0;  
    int visited[n];  
    for(int i=0; i<n; i++) visited[i] = 0;
```

```
    for(int i=0; i<n; i++) {  
        if(p[i].at == 0) {  
            readyQ[rear++] = i;  
            visited[i] = 1;  
        }  
    }
```

```
    while(completed < n) {  
        if(front == rear) {  
            time++;  
            for(int i=0; i<n; i++) {  
                if(!visited[i] && p[i].at <= time) {  
                    readyQ[rear++] = i;  
                    visited[i] = 1;  
                }  
            }
```

```

    }
    continue;
}

int idx = readyQ[front++];
if(rem_bt[idx] > tq) {
    time += tq;
    rem_bt[idx] -= tq;
    p[idx].rt = rem_bt[idx];
} else {
    time += rem_bt[idx];
    rem_bt[idx] = 0;
    p[idx].ct = time;
    p[idx].tat = p[idx].ct - p[idx].at;
    p[idx].wt = p[idx].tat - p[idx].bt;
    p[idx].rt = 0;
    completed++;
}

for(int i=0; i<n; i++) {
    if(!visited[i] && p[i].at <= time) {
        readyQ[rear++] = i;
        visited[i] = 1;
    }
}

if(rem_bt[idx] > 0) {
    readyQ[rear++] = idx;
}

}

printf("\nRRS scheduling:\n");
printf("PID\tAT\tBT\tCT\tTAT\tWT\n");
for(int i=0; i<n; i++) {
    printf("%d\t%d\t%d\t%d\t%d\t%d\t%d\n",
        p[i].pid, p[i].at, p[i].bt, p[i].ct,
        p[i].tat, p[i].wt);
}

float avg_TAT = 0, avg_WT = 0;
for(int i=0; i<n; i++) {
    avg_TAT += p[i].tat;
    avg_WT += p[i].wt;
}

printf("\nAverage turnaround time: %.2f ms\n", avg_TAT/n);
printf("Average waiting time: %.2f ms\n", avg_WT/n);

return 0;
}

```

```

Average WT = 10.00ms
mac@Prafuls-MacBook-Pro BMS % ./"RoundRobin"
Enter number of processes: 4
Enter Time Quantum: 2
Enter arrival times:
0
1
2
3
Enter burst times:
3
4
5
12

RRS scheduling:

```

PID	AT	BT	CT	TAT	WT
1	0	3	7	7	4
2	1	4	11	10	6
3	2	5	16	14	9
4	3	12	24	21	9

```

Average turnaround time: 13.00 ms
Average waiting time: 7.00 ms

```


PROGRAM 2

Question : Write a C program to simulate multi-level queue scheduling algorithm considering the following scenario. All the processes in the system are divided into two categories – system processes and user processes. System processes are to be given higher priority than user processes. Use FCFS scheduling for the processes in each queue.

SOLN :

```
#include <stdio.h>

#define MAX 10

typedef struct {
    int pid, at, bt, rt, ct, tat, wt, queue;
} Process;

int main() {
    Process p[MAX];
    int n, i, time = 0, completed = 0;
    int tq = 2;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    for (i = 0; i < n; i++) {
        printf("\nP%d\n", i + 1);
        p[i].pid = i + 1;

        printf("Arrival Time: ");
        scanf("%d", &p[i].at);

        printf("Burst Time: ");
        scanf("%d", &p[i].bt);

        printf("Queue (1-System, 2-User): ");
        scanf("%d", &p[i].queue);

        p[i].rt = p[i].bt;
    }

    printf("\nGantt Chart:\n");

    while (completed < n) {
        int executed = 0;

        for (i = 0; i < n; i++) {
            if (p[i].queue == 1 && p[i].rt > 0 && p[i].at <= time) {

                executed = 1;

                if (p[i].rt > tq) {
                    printf("P%d ", p[i].pid);
                    time += tq;
                }
            }
        }
    }
}
```

```

p[i].rt -= tq;
} else {
printf("P%d ", p[i].pid);
time += p[i].rt;
p[i].rt = 0;
p[i].ct = time;
completed++;
}
}
}

if (!executed) {
for (i = 0; i < n; i++) {
if (p[i].queue == 2 && p[i].rt > 0 && p[i].at <= time) {

executed = 1;

printf("P%d ", p[i].pid);
p[i].rt--;
time++;

if (p[i].rt == 0) {
p[i].ct = time;
completed++;
}

break;
}
}

if (!executed) {
time++;
}
}

float total_tat = 0, total_wt = 0;

printf("\n\nPID\tAT\tBT\tCT\tTAT\tWT\n");

for (i = 0; i < n; i++) {
p[i].tat = p[i].ct - p[i].at;
p[i].wt = p[i].tat - p[i].bt;

total_tat += p[i].tat;
total_wt += p[i].wt;

printf("P%d\t%d\t%d\t%d\t%d\t%d\n",
        p[i].pid, p[i].at, p[i].bt,
        p[i].ct, p[i].tat, p[i].wt);
}

printf("\nAverage TAT = %.2f", total_tat / n);
printf("\nAverage WT = %.2f\n", total_wt / n);

```

```

return 0;
}

```

```

● mac@Prafuls-MacBook-Pro SQL % gcc MultilevelQueue.c -o MultilevelQueue
● mac@Prafuls-MacBook-Pro SQL % ./MultilevelQueue
Enter number of processes: 4

P1
Arrival Time: 0
Burst Time: 4
Queue (1-System, 2-User): 1

P2
Arrival Time: 0
Burst Time: 3
Queue (1-System, 2-User): 1

P3
Arrival Time: 0
Burst Time: 8
Queue (1-System, 2-User): 2

P4
Arrival Time: 10
Burst Time: 5
Queue (1-System, 2-User): 1

Gantt Chart:
P1 P2 P1 P2 P3 P3 P3 P4 P4 P4 P3 P3 P3 P3 P3

PID   AT   BT   CT   TAT  WT
P1     0    4    6    6    2
P2     0    3    7    7    4
P3     0    8   20   20   12
P4    10    5   15    5    0

Average TAT = 9.50
Average WT = 4.50

```

PROGRAM 3

QUESTION : Write a C program to simulate Real-Time CPU Scheduling algorithms:

Rate- Monotonic

Earliest-deadline First

Proportional scheduling

SOLN : Rate Monotonic Scheduling

```
#include <stdio.h>
#include <stdlib.h>
#include <math.h>
struct Process {
    int pid;
    int burst;
    int period;
    int remaining;
};

int gcd(int a, int b) {
    if (b == 0) return a;
    return gcd(b, a % b);
}
int lcm(int a, int b) {
    return (a * b) / gcd(a, b);
}

int main() {
    int n;
    printf("Enter the number of processes: ");
    scanf("%d", &n);

    struct Process p[n];
    printf("Enter the CPU burst times:\n");
    for (int i = 0; i < n; i++) {
        scanf("%d", &p[i].burst);
        p[i].pid = i + 1;
    }

    printf("Enter the time periods:\n");
    int hyperPeriod = 1;
    for (int i = 0; i < n; i++) {
        scanf("%d", &p[i].period);
        hyperPeriod = lcm(hyperPeriod, p[i].period);
    }

    printf("\nLCM=%d\n", hyperPeriod);
```

```

printf("\nRate Monotone Scheduling:\n");
printf("PID\tBurst\tPeriod\n");
for (int i = 0; i < n; i++) {
    printf("%d\t%d\t%d\n", p[i].pid, p[i].burst, p[i].period);
}

double U = 0.0;
for (int i = 0; i < n; i++) {
    U += (double)p[i].burst / p[i].period;
}
double bound = n * (pow(2.0, 1.0/n) - 1);
printf("\n%.6f <= %.6f ==> %s\n", U, bound, (U <= bound ? "true" : "false"));
printf("Scheduling occurs for %d ms\n\n", hyperPeriod);

for (int t = 0; t < hyperPeriod; t++) {
    int chosen = -1;

    for (int i = 0; i < n; i++) {
        if (t % p[i].period == 0) {
            p[i].remaining = p[i].burst;
        }
    }

    for (int i = 0; i < n; i++) {
        if (p[i].remaining > 0) {
            chosen = i;
            break;
        }
    }

    if (chosen != -1) {
        printf("%dms onwards: Process %d running\n", t, p[chosen].pid);
        p[chosen].remaining--;
    } else {
        printf("%dms onwards: CPU is idle\n", t);
    }
}

return 0;
}

```

```
"C:\Users\BMSCECSE-F4\Desktop" X + v
Enter the number of processes: 2
Enter the CPU burst times:
20 35
Enter the time periods:
50 100
LCM=100

Rate Monotone Scheduling:
PID    Burst    Period
1       20       50
2       35      100

0.750000 <= 0.828427 =>true
Scheduling occurs for 100 ms

0ms onwards: Process 1 running
20ms onwards: Process 2 running
50ms onwards: Process 1 running
70ms onwards: Process 2 running
75ms onwards: CPU is idle

Process returned 0 (0x0)   execution time : 7.727 s
Press any key to continue.
|
```

SOLN : Earliest Deadline First

```
#include <stdio.h>
```

```
#include <limits.h>
```

```
#define MAX_TASKS 10
```

```
typedef struct {
```

```
    int pid;
```

```
    int burst;
```

```
    int deadline;
```

```
    int period;
```

```
} Task;
```

```
int gcd(int a, int b) {
```

```
    while (b != 0) {
```

```
        int temp = b;
```

```
        b = a % b;
```

```
        a = temp;
```

```
    }
```

```
    return a;
```

```
}
```

```
int lcm(int a, int b) {
```

```
    return (a * b) / gcd(a, b);
```

```
}
```

```
int main() {
```

```
    int n;
```

```
    Task tasks[MAX_TASKS];
```

```
    printf("Enter number of tasks: ");
```

```
    scanf("%d", &n);
```

```
    for (int i = 0; i < n; i++) {
```

```
        printf("\nTask %d:\n", i + 1);
```

```
        tasks[i].pid = i + 1;
```

```
        printf("Enter Burst Time: ");
```

```
        scanf("%d", &tasks[i].burst);
```

```
        printf("Enter Deadline: ");
```

```
        scanf("%d", &tasks[i].deadline);
```

```
        printf("Enter Period: ");
```

```
        scanf("%d", &tasks[i].period);
```

```
    }
```

```
    int SIMULATION_TIME = tasks[0].period;
```

```

for (int i = 1; i < n; i++) {
    SIMULATION_TIME = lcm(SIMULATION_TIME, tasks[i].period);
}

int remaining[MAX_TASKS] = {0};
int next_release[MAX_TASKS] = {0};
int abs_deadline[MAX_TASKS] = {0};
int active[MAX_TASKS] = {0};

printf("\nScheduling occurs for %d ms (LCM of periods)\n\n", SIMULATION_TIME);

for (int t = 0; t < SIMULATION_TIME; t++) {

    // Release jobs
    for (int i = 0; i < n; i++) {
        if (t == next_release[i]) {
            remaining[i] = tasks[i].burst;
            abs_deadline[i] = t + tasks[i].deadline;
            next_release[i] += tasks[i].period;
            active[i] = 1;
        }
    }

    int chosen = -1;
    int earliest = INT_MAX;

    for (int i = 0; i < n; i++) {
        if (active[i] && remaining[i] > 0) {
            if (abs_deadline[i] < earliest) {
                earliest = abs_deadline[i];
                chosen = i;
            }
        }
    }

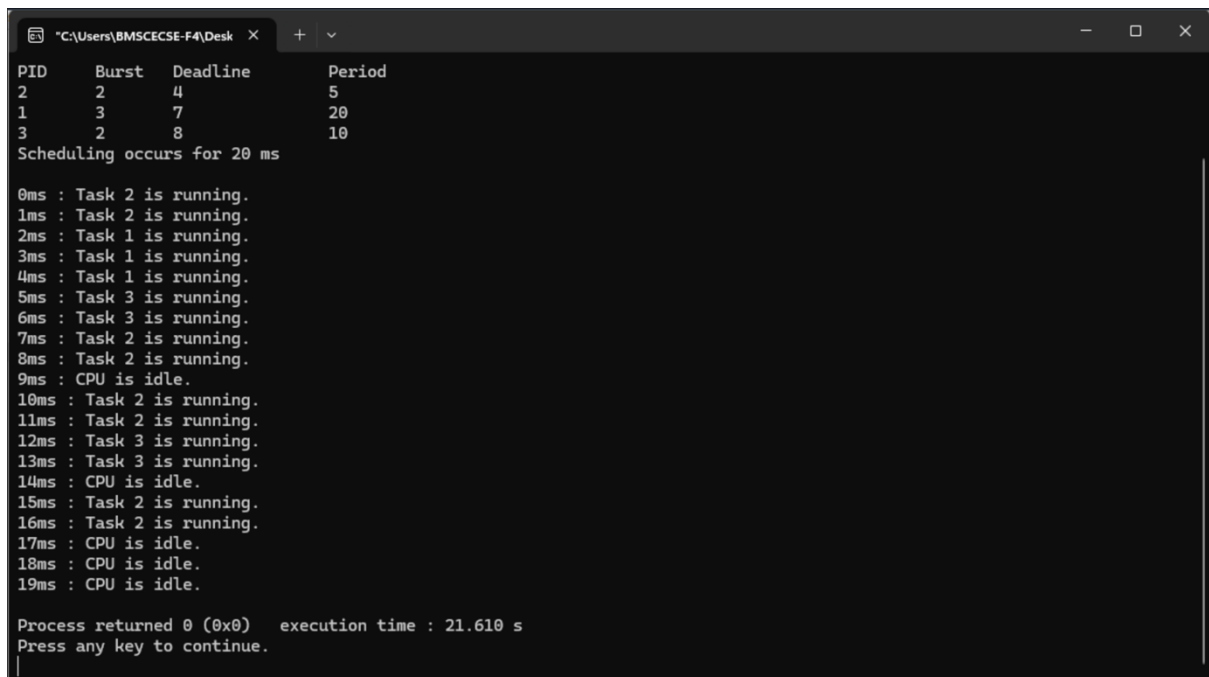
    if (chosen != -1) {
        printf("%dms : Task %d is running.\n", t, tasks[chosen].pid);
        remaining[chosen]--;

        if (remaining[chosen] == 0) {
            active[chosen] = 0;
        }
    } else {
        printf("%dms : CPU is idle.\n", t);
    }
}

```



```
    return 0;
}
```



The screenshot shows a Windows command prompt window with the title bar "C:\Users\BMSCECSE-F4\Desktop". The output displays a table of task parameters and a timeline of task execution.

PID	Burst	Deadline	Period
2	2	4	5
1	3	7	20
3	2	8	10

Scheduling occurs for 20 ms

0ms : Task 2 is running.
1ms : Task 2 is running.
2ms : Task 1 is running.
3ms : Task 1 is running.
4ms : Task 1 is running.
5ms : Task 3 is running.
6ms : Task 3 is running.
7ms : Task 2 is running.
8ms : Task 2 is running.
9ms : CPU is idle.
10ms : Task 2 is running.
11ms : Task 2 is running.
12ms : Task 3 is running.
13ms : Task 3 is running.
14ms : CPU is idle.
15ms : Task 2 is running.
16ms : Task 2 is running.
17ms : CPU is idle.
18ms : CPU is idle.
19ms : CPU is idle.

Process returned 0 (0x0) execution time : 21.610 s
Press any key to continue.

SOLN : Proportional scheduling

PROGRAM 4 :

QUESTION :

- 4a. Write a C program to simulate producer-consumer problem using semaphores
- 4b. Write a C program to simulate the concept of Dining Philosophers problem

SOLN : Bounded Buffer

```
#include <stdio.h>
#include <stdlib.h>
#define BUFFER_SIZE 5

int buffer[BUFFER_SIZE];
int in = 0;
int out = 0;

int empty = BUFFER_SIZE;
int full = 0;

void produce() {
    if (empty == 0) {
        printf("\nBuffer is full! Cannot produce.\n");
        return;
    }

    int item;
    printf("Enter item to produce: ");
    scanf("%d", &item);

    buffer[in] = item;
    printf("Produced: %d\n", item, in);

    in = (in + 1) % BUFFER_SIZE;
    empty--;
    full++;
}

void consume() {
    if (full == 0) {
        printf("Buffer is empty! Cannot consume.\n");
        return;
    }

    int item = buffer[out];
    printf("Consumed: %d\n", item, out);

    out = (out + 1) % BUFFER_SIZE;
```

```

    full--;
    empty++;
}

int main() {
    int choice;
    printf("Buffer Size: %d\n", BUFFER_SIZE);

    while (1) {
        printf("\n1. Produce\n2. Consume\n3. Exit\n");
        printf("Current State: [Full slots: %d | Empty slots: %d]\n", full, empty);
        printf("Enter choice: ");

        if (scanf("%d", &choice) != 1) break;

        switch (choice) {
            case 1:
                produce();
                break;
            case 2:
                consume();
                break;
            case 3:
                printf("Exiting...\n");
                exit(0);
            default:
                printf("Invalid choice!\n");
        }
    }

    return 0;
}

```

```

mac@Prafuls-MacBook-Pro SQL % /Users/mac/Desktop/SQL/BoundedBuffer
1. Produce
2. Consume
3. Exit
Enter choice: 1
Enter item to produce: 12
Producer produced: 12 at position 0

1. Produce
2. Consume
3. Exit
Enter choice: 2
Consumer consumed: 12 from position 0

1. Produce
2. Consume
3. Exit
Enter choice: 3
Exiting. A Praful Srinivasan

```

SOLN : Dining Philosopher

```
#include<stdio.h>
#include<pthread.h>
#include<unistd.h>
#define N 5
pthread_mutex_t forks[N];
pthread_t philosophers[N];

void* philosopher(void* num)
{
    int id = *(int*)num;
    int left = id; // left fork index
    int right = (id + 1) % N; // right fork index
    while (1)
    {
        printf("Philosopher %d is thinking.\n", id);
        sleep(1); // Thinking
        // To avoid deadlock, philosophers with even id pick left then right,
        // odd id pick right then left.
        if (id % 2 == 0)
        {
            pthread_mutex_lock(&forks[left]);
            printf("Philosopher %d picked up left fork %d.\n", id, left);
            pthread_mutex_lock(&forks[right]);
            printf("Philosopher %d picked up right fork %d.\n", id, right); }
        else
        {
            pthread_mutex_lock(&forks[right]);
            printf("Philosopher %d picked up right fork %d.\n", id, right); // Pick up left
            pthread_mutex_lock(&forks[left]);
            printf("Philosopher %d picked up left fork %d.\n", id, left);
        }
        printf("Philosopher %d is eating.\n", id);
        sleep(2); // Put down forks
        pthread_mutex_unlock(&forks[left]);
        pthread_mutex_unlock(&forks[right]);
        printf("Philosopher %d put down forks %d and %d.\n", id, left, right);
    }
    return NULL;
}

int main() {
    int i;
    int ids[N]; // Initialize forks (mutexes)
    for (i = 0; i < N; i++)
    { pthread_mutex_init(&forks[i], NULL);
```

```

ids[i] = i;
}
for (i = 0; i < N; i++){
pthread_create(&philosophers[i], NULL, philosopher, &ids[i]);
}

for (i = 0; i < N; i++){
pthread_join(philosophers[i], NULL); }
for (i = 0; i < N; i++){
pthread_mutex_destroy(&forks[i]);
}
return 0;
}

```

```

mac@Prafuls-MacBook-Pro SQL % /Users/mac/Desktop/SQL/BoundedBuffer
Philosopher 0 is thinking.
Philosopher 4 is thinking.
Philosopher 1 is thinking.
Philosopher 2 is thinking.
Philosopher 3 is thinking.
Philosopher 0 picked up left fork 0.
Philosopher 0 picked up right fork 1.
Philosopher 0 is eating.
Philosopher 4 picked up left fork 4.
Philosopher 1 picked up right fork 2.
Philosopher 0 put down forks 0 and 1.
Philosopher 0 is thinking.
Philosopher 4 picked up right fork 0.
Philosopher 4 is eating.
Philosopher 1 picked up left fork 1.
Philosopher 1 is eating.
Philosopher 4 put down forks 4 and 0.
Philosopher 4 is thinking.
Philosopher 3 picked up right fork 4.
Philosopher 2 picked up left fork 2.
Philosopher 0 picked up left fork 0.
Philosopher 0 picked up right fork 1.
Philosopher 0 is eating.
Philosopher 3 picked up left fork 3.
Philosopher 3 is eating.
Philosopher 1 put down forks 1 and 2.
Philosopher 1 is thinking.
Philosopher 0 put down forks 0 and 1.
Philosopher 4 picked up left fork 4.
Philosopher 4 picked up right fork 0.
Philosopher 2 picked up right fork 3.
Philosopher 2 is eating.
Philosopher 4 is eating.
Philosopher 0 is thinking.
Philosopher 3 put down forks 3 and 4.
Philosopher 3 is thinking.
Philosopher 2 put down forks 2 and 3.
Philosopher 2 is thinking.
Philosopher 4 put down forks 4 and 0.
Philosopher 4 is thinking.
Philosopher 1 picked up right fork 2.
Philosopher 0 picked up left fork 0.
Philosopher 1 picked up left fork 1.
Philosopher 1 is eating.
Philosopher 3 picked up right fork 4.
Philosopher 3 picked up left fork 3.
Philosopher 3 is eating.
Philosopher 3 put down forks 3 and 4.

```

PROGRAM 5

QUESTION : Write a C program to simulate Bankers algorithm and deadlock avoidance

SOLN : Banker algorithm

```
#include <stdio.h>
#include <stdbool.h>

#define MAX_PROCESSES 10
#define MAX_RESOURCES 10

int main() {
    int n, m;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter number of resource types: ");
    scanf("%d", &m);

    int allocation[MAX_PROCESSES][MAX_RESOURCES];
    int max[MAX_PROCESSES][MAX_RESOURCES];
    int need[MAX_PROCESSES][MAX_RESOURCES];
    int available[MAX_RESOURCES];

    printf("\nEnter Allocation Matrix:\n");
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < m; j++) {
            scanf("%d", &allocation[i][j]);
        }
    }

    printf("\nEnter Max Matrix:\n");
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < m; j++) {
            scanf("%d", &max[i][j]);
        }
    }

    printf("\nEnter Available Resources:\n");
    for (int j = 0; j < m; j++) {
        scanf("%d", &available[j]);
    }

    for (int i = 0; i < n; i++) {
        for (int j = 0; j < m; j++) {
```

```

        need[i][j] = max[i][j] - allocation[i][j];
    }
}

bool finish[MAX_PROCESSES] = {false};
int safeSequence[MAX_PROCESSES];

int work[MAX_RESOURCES];

// Work = Available
for (int j = 0; j < m; j++) {
    work[j] = available[j];
}

int count = 0;

while (count < n) {
    bool found = false;

    for (int i = 0; i < n; i++) {

        if (finish[i] == false) {

            bool canExecute = true;

            for (int j = 0; j < m; j++) {
                if (need[i][j] > work[j]) {
                    canExecute = false;
                    break;
                }
            }

            if (canExecute) {

                for (int j = 0; j < m; j++) {
                    work[j] += allocation[i][j];
                }

                safeSequence[count] = i;
                count++;

                finish[i] = true;
                found = true;
            }
        }
    }
}

```



```

        if (!found) {
            break;
        }
    }

    if (count == n) {
        printf("\nSystem is in SAFE state.\n");
        printf("Safe Sequence: ");

        for (int i = 0; i < n; i++) {
            printf("P%d", safeSequence[i]);

            if (i != n - 1) {
                printf(" -> ");
            }
        }

        printf("\n");
    }
    else {
        printf("\nSystem is NOT in safe state.\n");
    }

    return 0;
}

```

```

mac@Prafuls-MacBook-Pro SQL % /Users/mac/Desktop/SQL/Bankers
Enter number of processes: 5
Enter number of resource types: 3

Enter Allocation Matrix:
0 1 0
2 0 0
3 0 2
2 1 1
0 0 2

Enter Max Matrix:
7 5 3
3 2 2
9 0 2
2 2 2
4 3 3

Enter Available Resources:
3 3 2

System is in SAFE state.
Safe Sequence: P1 -> P3 -> P4 -> P0 -> P2

```

SOLN : Deadlock avoidance algorithm

```
#include <stdio.h>
#include <stdbool.h>

#define MAX_PROCESSES 10
#define MAX_RESOURCES 10

int main() {
    int n, m;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter number of resource types: ");
    scanf("%d", &m);

    int allocation[MAX_PROCESSES][MAX_RESOURCES];
    int request[MAX_PROCESSES][MAX_RESOURCES];
    int available[MAX_RESOURCES];

    printf("\nEnter Allocation Matrix:\n");
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < m; j++) {
            scanf("%d", &allocation[i][j]);
        }
    }

    printf("\nEnter Request Matrix:\n");
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < m; j++) {
            scanf("%d", &request[i][j]);
        }
    }

    printf("\nEnter Available Resources:\n");
    for (int j = 0; j < m; j++) {
        scanf("%d", &available[j]);
    }

    bool finish[MAX_PROCESSES];
    int work[MAX_RESOURCES];
    int safeSequence[MAX_PROCESSES];

    for (int j = 0; j < m; j++) {
        work[j] = available[j];
    }
```

```

for (int i = 0; i < n; i++) {
    bool zeroAllocation = true;

    for (int j = 0; j < m; j++) {
        if (allocation[i][j] != 0) {
            zeroAllocation = false;
            break;
        }
    }

    finish[i] = zeroAllocation;
}

int count = 0;

while (1) {
    bool found = false;

    for (int i = 0; i < n; i++) {

        if (finish[i] == false) {

            bool canExecute = true;

            for (int j = 0; j < m; j++) {
                if (request[i][j] > work[j]) {
                    canExecute = false;
                    break;
                }
            }

            if (canExecute) {

                for (int j = 0; j < m; j++) {
                    work[j] += allocation[i][j];
                }

                finish[i] = true;
                safeSequence[count] = i;
                count++;

                found = true;
            }
        }
    }
}

```

```

        if (!found) {
            break;
        }
    }

    bool deadlock = false;

    for (int i = 0; i < n; i++) {
        if (finish[i] == false) {
            deadlock = true;
            break;
        }
    }

    if (!deadlock) {
        printf("\nNo Deadlock detected.\n");

        printf("Safe Sequence: ");
        for (int i = 0; i < count; i++) {
            printf("P%d", safeSequence[i]);

            if (i != count - 1) {
                printf(" -> ");
            }
        }

        printf("\n");
    }
    else {
        printf("\nDeadlock detected.\n");

        printf("Processes involved in deadlock: ");

        for (int i = 0; i < n; i++) {
            if (finish[i] == false) {
                printf("P%d ", i);
            }
        }

        printf("\n");
    }

    return 0;
}

```

```
mac@Pratuls-MacBook-Pro SQL % /Users/mac/Desktop/SQL/Deadlock
Enter number of resource types: 3

Enter Allocation Matrix:
0 1 0
2 0 0
3 0 3

Enter Request Matrix:
0 0 0
2 0 2
0 0 1

Enter Available Resources:
0 0 0

Deadlock detected.
Processes involved in deadlock: P1 P2
mac@Pratuls-MacBook-Pro SQL %
mac@Pratuls-MacBook-Pro SQL % /Users/mac/Desktop/SQL/Deadlock
Enter number of processes: 3
Enter number of resource types: 3

Enter Allocation Matrix:
0 1 0
2 0 0
3 0 3

Enter Request Matrix:
0 0 0
2 0 2
0 0 0

Enter Available Resources:
0 0 0

No Deadlock detected.
Safe Sequence: P0 -> P2 -> P1
```

PROGRAM 6 :

QUESTION : Write a C program to simulate Memory Allocation (First Fit, Best Fit, Worst Fit)

SOLN : Memory Allocation

```
#include <stdio.h>

void firstFit(int blockSize[], int m, int processSize[], int n) {
    int allocation[n];

    for (int i = 0; i < n; i++)
        allocation[i] = -1;

    for (int i = 0; i < n; i++) {
        for (int j = 0; j < m; j++) {
            if (blockSize[j] >= processSize[i]) {
                allocation[i] = j;
                blockSize[j] = 0;
                break;
            }
        }
    }

    printf("\nFirst Fit Allocation:\n");
    printf("Process No.\tProcess Size\tBlock No.\n");

    for (int i = 0; i < n; i++) {
        printf("%d\t\t%d\t\t", i + 1, processSize[i]);

        if (allocation[i] != -1)
            printf("%d\n", allocation[i] + 1);
        else
            printf("Not Allocated\n");
    }
}

void bestFit(int blockSize[], int m, int processSize[], int n) {
    int allocation[n];

    for (int i = 0; i < n; i++)
        allocation[i] = -1;

    for (int i = 0; i < n; i++) {
        int bestIdx = -1;

        for (int j = 0; j < m; j++) {
```

```

        if (blockSize[j] >= processSize[i]) {
            if (bestIdx == -1 || blockSize[j] < blockSize[bestIdx]) {
                bestIdx = j;
            }
        }
    }

    if (bestIdx != -1) {
        allocation[i] = bestIdx;
        blockSize[bestIdx] = 0;
    }
}

printf("\nBest Fit Allocation:\n");
printf("Process No.\tProcess Size\tBlock No.\n");

for (int i = 0; i < n; i++) {
    printf("%d\t%d\t", i + 1, processSize[i]);

    if (allocation[i] != -1)
        printf("%d\n", allocation[i] + 1);
    else
        printf("Not Allocated\n");
}

}

void worstFit(int blockSize[], int m, int processSize[], int n) {
    int allocation[n];

    for (int i = 0; i < n; i++)
        allocation[i] = -1;

    for (int i = 0; i < n; i++) {
        int worstIdx = -1;

        for (int j = 0; j < m; j++) {
            if (blockSize[j] >= processSize[i]) {
                if (worstIdx == -1 || blockSize[j] > blockSize[worstIdx]) {
                    worstIdx = j;
                }
            }
        }

        if (worstIdx != -1) {
            allocation[i] = worstIdx;
            blockSize[worstIdx] = 0;
        }
    }
}

```

```

    }

    printf("\nWorst Fit Allocation:\n");
    printf("Process No.\tProcess Size\tBlock No.\n");

    for (int i = 0; i < n; i++) {
        printf("%d\t%d\t", i + 1, processSize[i]);

        if (allocation[i] != -1)
            printf("%d\n", allocation[i] + 1);
        else
            printf("Not Allocated\n");
    }
}

int main() {
    int m, n;

    printf("Enter number of memory blocks: ");
    scanf("%d", &m);

    int blocks[m], firstBlocks[m], bestBlocks[m], worstBlocks[m];

    printf("Enter sizes of memory blocks:\n");
    for (int i = 0; i < m; i++) {
        scanf("%d", &blocks[i]);
        firstBlocks[i] = blocks[i];
        bestBlocks[i] = blocks[i];
        worstBlocks[i] = blocks[i];
    }

    printf("Enter number of processes: ");
    scanf("%d", &n);

    int processSize[n];

    printf("Enter sizes of processes:\n");
    for (int i = 0; i < n; i++) {
        scanf("%d", &processSize[i]);
    }

    firstFit(firstBlocks, m, processSize, n);
    bestFit(bestBlocks, m, processSize, n);
    worstFit(worstBlocks, m, processSize, n);

    return 0;
}

```



```

mac@Prafuls-MacBook-Pro SQL % "/Users/mac/Desktop/SQL/ LAB 6"
Enter number of memory blocks: 5
Enter sizes of memory blocks:
400 700 200 300 600
Enter number of processes: 4
Enter sizes of processes:
212 512 312 516

--- First Fit ---
Process No.    Process Size    Block No.
1              212          1
2              512          2
3              312          5
4              516        Not Allocated

--- Best Fit ---
Process No.    Process Size    Block No.
1              212          4
2              512          5
3              312          1
4              516          2

--- Worst Fit ---
Process No.    Process Size    Block No.
1              212          2
2              512          5
3              312          1
4              516        Not Allocated

```

PROGRAM 7 :

QUESTION : Write a C program to simulate Page Replacement

```
#include <stdio.h>
#include <stdbool.h>

#define MAX_PAGES 100

void printFrames(int frames[], int num_frames) {
    printf("[");
    int count = 0;
    for (int i = 0; i < num_frames; i++) {
        if (frames[i] != -1) {
            printf("%d", frames[i]);
            count++;
        }
    }
    printf("]\n");
}

void fifoPageReplacement(int pages[], int num_pages, int num_frames) {
    printf("\n--- FIFO Page Replacement ---\n");

    int frames[num_frames];
    for (int i = 0; i < num_frames; i++) frames[i] = -1;

    int page_faults = 0;
    int next = 0;

    for (int i = 0; i < num_pages; i++) {
        int page = pages[i];
        bool is_present = false;

        for (int j = 0; j < num_frames; j++) {
            if (frames[j] == page) {
                is_present = true;
                break;
            }
        }

        printf("Page %d -> ", page);

        if (!is_present) {
```

```

        frames[next] = page;
        next = (next + 1) % num_frames;
        page_faults++;
    }

    printFrames(frames, num_frames);
}
printf("Total Page Faults (FIFO): %d\n", page_faults);
}

void lruPageReplacement(int pages[], int num_pages, int num_frames) {
    printf("\n--- LRU Page Replacement ---\n");

    int frames[num_frames];
    int last_used[num_frames];
    for (int i = 0; i < num_frames; i++) {
        frames[i] = -1;
        last_used[i] = -1;
    }

    int page_faults = 0;

    for (int i = 0; i < num_pages; i++) {
        int page = pages[i];
        bool is_present = false;
        int present_index = -1;

        for (int j = 0; j < num_frames; j++) {
            if (frames[j] == page) {
                is_present = true;
                present_index = j;
                break;
            }
        }

        printf("Page %d -> ", page);

        if (is_present) {

            last_used[present_index] = i;
        } else {

            page_faults++;

```

```

    int empty_index = -1;
    for (int j = 0; j < num_frames; j++) {
        if (frames[j] == -1) {
            empty_index = j;
            break;
        }
    }

    if (empty_index != -1) {

        frames[empty_index] = page;
        last_used[empty_index] = i;
    } else {

        int lru_index = 0;
        int min_time = last_used[0];
        for (int j = 1; j < num_frames; j++) {
            if (last_used[j] < min_time) {
                min_time = last_used[j];
                lru_index = j;
            }
        }
        frames[lru_index] = page;
        last_used[lru_index] = i;
    }
}

printFrames(frames, num_frames);
}
printf("Total Page Faults (LRU): %d\n", page_faults);
}

```

```

void optimalPageReplacement(int pages[], int num_pages, int num_frames) {
    printf("\n--- Optimal Page Replacement ---\n");

    int frames[num_frames];
    for (int i = 0; i < num_frames; i++) frames[i] = -1;

    int page_faults = 0;

    for (int i = 0; i < num_pages; i++) {
        int page = pages[i];
        bool is_present = false;

        for (int j = 0; j < num_frames; j++) {
            if (frames[j] == page) {

```

```

        is_present = true;
        break;
    }
}

printf("Page %d -> ", page);

if (!is_present) {
    page_faults++;

    int empty_index = -1;
    for (int j = 0; j < num_frames; j++) {
        if (frames[j] == -1) {
            empty_index = j;
            break;
        }
    }

    if (empty_index != -1) {
        frames[empty_index] = page;
    } else {

        int replace_index = -1;
        int farthest_use = -1;

        for (int j = 0; j < num_frames; j++) {
            int next_use = -1;

            for (int k = i + 1; k < num_pages; k++) {
                if (pages[k] == frames[j]) {
                    next_use = k;
                    break;
                }
            }

            if (next_use == -1) {
                replace_index = j;
                break;
            }

            if (next_use > farthest_use) {
                farthest_use = next_use;
                replace_index = j;
            }
        }
    }
}

```

```

        frames[replace_index] = page;
    }
}

printFrames(frames, num_frames);
}
printf("Total Page Faults (Optimal): %d\n", page_faults);
}

int main() {
    int num_pages, num_frames;
    int pages[MAX_PAGES];

    printf("Enter number of pages: ");
    if (scanf("%d", &num_pages) != 1 || num_pages <= 0 || num_pages > MAX_PAGES) {
        printf("Invalid number of pages.\n");
        return 1;
    }

    printf("Enter page reference string:\n");
    for (int i = 0; i < num_pages; i++) {
        scanf("%d", &pages[i]);
    }

    printf("Enter number of frames: ");
    if (scanf("%d", &num_frames) != 1 || num_frames <= 0) {
        printf("Invalid number of frames.\n");
        return 1;
    }

    fifoPageReplacement(pages, num_pages, num_frames);
    lruPageReplacement(pages, num_pages, num_frames);
    optimalPageReplacement(pages, num_pages, num_frames);

    return 0;
}

```

```

Enter number of pages: 21
Enter page reference string:
5
0
1
0
2
3
0
2
4
3
3
2
0
2
1
2
7
0
1
1
0
Enter number of frames: 3

--- FIFO Page Replacement ---
Page 5 -> [5]
Page 0 -> [50]
Page 1 -> [501]
Page 0 -> [501]
Page 2 -> [201]
Page 3 -> [211]
Page 0 -> [230]
Page 2 -> [230]
Page 4 -> [430]
Page 3 -> [430]
Page 3 -> [430]
Page 2 -> [420]
Page 0 -> [420]
Page 2 -> [420]
Page 1 -> [421]
Page 2 -> [421]
Page 7 -> [721]
Page 0 -> [701]
Page 1 -> [701]
Page 1 -> [701]
Page 0 -> [701]
Total Page Faults (FIFO): 11

--- LRU Page Replacement ---
Page 5 -> [5]
Page 0 -> [50]

```

```

--- LRU Page Replacement ---
Page 5 -> [5]
Page 0 -> [50]
Page 1 -> [501]
Page 0 -> [501]
Page 2 -> [201]
Page 3 -> [203]
Page 0 -> [203]
Page 2 -> [203]
Page 4 -> [204]
Page 3 -> [230]
Page 3 -> [230]
Page 2 -> [230]
Page 0 -> [230]
Page 2 -> [230]
Page 1 -> [210]
Page 2 -> [210]
Page 7 -> [217]
Page 0 -> [207]
Page 1 -> [107]
Page 1 -> [107]
Page 0 -> [107]
Total Page Faults (LRU): 12

--- Optimal Page Replacement ---
Page 5 -> [5]
Page 0 -> [50]
Page 1 -> [501]
Page 0 -> [501]
Page 2 -> [201]
Page 3 -> [203]
Page 0 -> [203]
Page 2 -> [203]
Page 4 -> [203]
Page 3 -> [203]
Page 3 -> [203]
Page 2 -> [203]
Page 0 -> [203]
Page 2 -> [203]
Page 1 -> [201]
Page 2 -> [201]
Page 7 -> [701]
Page 0 -> [701]
Page 1 -> [701]
Page 1 -> [701]
Page 0 -> [701]
Total Page Faults (Optimal): 9

```